

IPA Data Science Assessment

Thank you for your interest in Data Science at Innovations for Poverty Action!

Context

For this assessment, you will be working with a dataset of survey responses from a randomized control trial (RCT) conducted by Innovations for Poverty Action in Ghana.

The RCT is described in the paper provided, “The Impact of Financial Education for Youth in Ghana” (2018), by James Berry, Dean Karlan, and Menno Pradhan in *World Development* 102(2018) 71-89.

The study evaluates the impact of two financial education programs, Aflatoun and Honest Money Box (HMB), on youth savings behavior in Ghana. Aflatoun is a non-governmental organization that developed school-based financial literacy education accompanied by social reinforcement through school savings clubs. HMB encompasses a similar set of financial literacy education without the social component of the Aflatoun program.

The study was conducted in 125 primary and junior high schools in Ghana in southern and eastern Ghana in 2010 and 2011. 45 schools were randomly assigned to participate in the full Aflatoun program, 45 schools were randomly assigned to the Honest Money Box program, and 45 schools were randomly assigned to the control group. Random assignment was stratified by region and average class size of schools.

Data

Use the dataset provided, `ghana_youth_savings.csv`, for your analysis. These data are a subset of the full survey data available via Berry, James; Karlan, Dean; Pradhan, Menno, 2020, “The Impact of Financial Education for Youth in Ghana”, <https://doi.org/10.7910/DVN/GDP2YP>, Harvard Dataverse, V1.

Each row in the dataset represents a unique student from a participant in the RCT. Data are from a panel survey with two rounds of interviews. All students were interviewed at baseline and then again at endline. If a student completed the endline survey, this is indicated in the `noattrit` column in the data.

In the data, the `aflatoun` and `hmb` columns indicate whether the school was randomly assigned to one of the two treatment groups. The `treatment` column indicates whether the school was assigned to either the Aflatoun or Honest Money Box program.

The columns in the dataset are as follows:

Column Name	Description
uuid	Student unique identifier
schid	School unique identifier
noattrit	Follow-up survey completed
aflatoun	School randomly assigned to Aflatoun program
hmb	School randomly assigned to Honesty Money Box program
treatment	School randomly assigned to either Aflatoun or HMB program
class	Student grade level at baseline
std_avg_class_size	Standardized verage class size at baseline
region1	Indicator of Region 1
region2	Indicator of Region 2
region3	Indicator of Region 3
urban	Student lives in urban area
gender	Gender of student
female	Student is female
age	Age of student at baseline
tribe	Ethnic tribe of student
hhsiz	Number of people in student's household at baseline
hhearn	Number of income earners in student's household at baseline
hhbankaccounts	Number of bank accounts in student's household at baseline
repeated_grade	Student ever repeated a grade at time of baseline
baseline_save	Student has money saved at time of baseline
baseline_saveamt	Amount of money saved at time of baseline
takeupschool	School participated in Aflatoun or Honesty Money Box program
takeup	Student participated in Aflatoun or Honesty Money Box program
end_save	Student has money saved at time of endline
end_saveamt	Amount of money saved at time of endline
end_saveschl	Student has money saved at school at time of endline
end_saveschlamt	Amount of money saved at school at time of endline
asset_index	Index of durable goods ownership at time of baseline
work_index	Index student's work participation at time of baseline

Column Name	Description
riskpref_index	Index of student's risk preferences at time of baseline
timepref_index	Index of student's time preferences at time of baseline
saving_attitude_index	Index of student's saving attitudes at time of baseline
saving_env_index	Index of home saving support at time of baseline
fin_lit_index	Index of financial literacy at time of baseline
tempt_index	Index of expenditure temptations at time of baseline
expend_index	Index of student's expenditure on self at time of baseline
academic_index	Index of student's academic performance at time of baseline

Instructions

For this assessment, **use Python to complete the tasks described below**. You may use any libraries you like, but please provide a file that describes how to reproduce your Python environment (e.g. a `pyproject.toml` or `requirements.txt`) file with the libraries you used.

Don't worry about the output matching the paper exactly, your main focus should be on writing code that is clear, easy to understand, and addresses the tasks described below.

You may use AI tools to assist with writing your code. However, you should declare what tools you use. And be prepare to explain the code as well as any improvements you would make if you had more time to work on the project.

Please submit your code and any other relevant files to the email address provided.

Limit yourself to 4 hours to complete the tasks. If you are unable to complete within the time limit, please submit what you have completed.

Tasks

1. Create a table that shows baseline summary statistics among the control observations for the variables shown in Column 1 of Table 3, "Control mean" of the paper. This table should include the mean, standard deviation of each of the variables in the table.

Bonus: Create the table with Columns 2 and 3 showing the difference between the control mean and the treatment means for students in Aflatoun and Honest Money Box (HMB) schools.

Table 3
Baseline summary statistics and orthogonality tests

	Control mean (1)	Difference from control		p-Value from F-test		Obs. (6)
		Aflatoun (2)	HMB (3)	Afla = HMB (4)	Afla = HMB = Control (5)	
Female	0.500 [0.500]	−0.003 (0.020)	−0.020 (0.019)	0.395	0.525	5291
Age	12.806 [1.989]	0.256 (0.204)	−0.058 (0.219)	0.126	0.250	5359
Student Has Money Saved	0.467 [0.499]	−0.040 (0.027)	−0.005 (0.028)	0.223	0.275	5362
Amount Saved	5.041 [17.966]	−0.792 (0.605)	−0.236 (0.846)	0.480	0.401	5337
Savings Attitudes Index	0.000 [1.000]	0.076 (0.060)	0.004 (0.057)	0.232	0.375	5362
Home Savings Support Index	0.000 [1.000]	−0.013 (0.047)	0.023 (0.047)	0.461	0.755	5364
Work Index	0.000 [1.000]	−0.102 (0.061)	0.014 (0.065)	0.0332	0.0678	5364
Risk Preference Index	0.000 [1.000]	0.036 (0.056)	0.073 (0.059)	0.514	0.469	5354
Time Preference Index	0.000 [1.000]	0.011 (0.049)	0.008 (0.054)	0.956	0.975	5355
Financial Literacy Index	0.000 [1.000]	0.052 (0.071)	0.016 (0.074)	0.571	0.726	5364
Expenditures on Temptation Goods Index	0.000 [1.000]	−0.113** (0.047)	−0.047 (0.058)	0.237	0.0574	5364
Expenditures on Self Index	0.000 [1.000]	0.004 (0.045)	0.053 (0.095)	0.599	0.854	5352
Academic Index	0.000 [1.000]	0.015 (0.077)	−0.067 (0.088)	0.313	0.589	5364
Completed Endline Survey	0.986 [0.116]	0.000 (0.004)	0.000 (0.004)	0.905	0.993	5364

Columns (2)–(5) present the results of regressions of the variable in each row on Aflatoun and HMB treatment dummies, controlling for stratification dummies (region and standardized average class size). Standard deviations in square brackets; standard errors, clustered at the school level, in parentheses. ** $p < 0.01$, * $p < 0.05$, $p < 0.1$.

2. Create a graph that demonstrates the difference in whether or not a student saved money at school at endline between the control group and the treatment group.

Bonus: Create a graph that demonstrates the difference in the amount saved at school at endline between the control group and the treatment group.

3. Using a regression model, estimate the causal impact of the Aflatoun and Honest Money Box programs on whether the student has save money at the time of endline and the amount saved at endline. For reference, the analysis should resemble the first two outcomes listed in Table 5. (For the purposes of this exercise, you can ignore multiple hypothesis correction, FDR-corrected q-values, mentioned in the paper.) Refer to the notes in Table 5 for the regression specification:

Columns (1) and (2) present individual regressions of each variable or index on Aflatoun and HMB treatment dummies. Column (4) presents individual regressions of each variable or index on dummies for the pooled treatment. Regressions control for stratification dummies (region and standardized average class size) and baseline values for the index, if available.

Table 5
Treatment effects on key outcome variables

Outcome variable	Aflatoun (1)	Honest money box (2)	p-Value, Afla = HMB (3)	Pooled effect (4)	Obs (5)
Student Has Money Saved	0.024 (0.022) [0.824]	0.031 (0.025) [0.824]	0.769	0.027 (0.020) [0.656]	5291
Amount Saved	-0.287 (1.297) [0.861]	-1.088 (1.264) [0.836]	0.501	-0.686 (1.136) [0.656]	5291
Savings Attitudes Index (positive attitude toward savings)	0.013 (0.043) [0.861]	0.049 (0.048) [0.824]	0.468	0.031 (0.039) [0.656]	5291
Home Savings Support Index (home environment conducive to saving)	-0.027 (0.050) [0.836]	0.052 (0.049) [0.824]	0.134	0.012 (0.042) [0.841]	5291
Work Index (more likely to work, more hours, etc.)	0.038 (0.049) [0.836]	0.102 (0.056) [0.824]	0.257	0.070 (0.045) [0.656]	5291
Risk Preference Index (higher = less risk averse)	-0.064 (0.054) [0.824]	-0.076 (0.054) [0.824]	0.804	-0.070 (0.049) [0.656]	5291
Time Preference Index (higher = lower discount rate of the future)	0.033 (0.049) [0.836]	0.031 (0.052) [0.836]	0.975	0.032 (0.043) [0.656]	5291
Financial Literacy Index	0.015 (0.055) [0.861]	-0.005 (0.057) [0.929]	0.714	0.005 (0.049) [0.915]	5291
Expenditures on Temptation Goods Index (propensity to spend on temptation goods)	-0.033 (0.048) [0.836]	-0.022 (0.044) [0.836]	0.766	-0.027 (0.042) [0.656]	5291
Expenditures on Self Index (propensity to spend on goods for self)	-0.016 (0.050) [0.861]	-0.064 (0.046) [0.824]	0.287	-0.040 (0.043) [0.656]	5291
Confidence Index	-0.047 (0.045) [0.824]	-0.011 (0.045) [0.861]	0.456	-0.029 (0.038) [0.656]	5291
Academic Performance Index (school attendance and test scores)	-0.033 (0.064) [0.836]	-0.047 (0.064) [0.836]	0.798	-0.040 (0.058) [0.656]	5291

Columns (1) and (2) present individual regressions of each variable or index on Aflatoun and HMB treatment dummies. Column (4) presents individual regressions of each variable or index on dummies for the pooled treatment. Regressions control for stratification dummies (region and standardized average class size) and baseline values for the index, if available. Missing values of baseline variables are coded as zero, and additional dummy variables are included to indicate missing values. Indices are aggregated ignoring missing values in the individual variables. See Appendix Tables 8–17 for component variables of each index. Money amounts reported in Ghana cedis. Standard errors clustered at the school level, in parentheses; FDR-corrected q-values in square brackets using procedure described in [Benjamini and Hochberg \(1995\)](#). For calculation of q-values, coefficients are grouped by specification (all coefficients in Columns (1) and (2) or all coefficients in Column (4)). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

4. In the paper, the author use a linear regression to predict whether or not the student participates in either the Aflatoun or Honest Money Box program. The linear regression as implemented by the authors is described in Table 4.

Using the variable, takeup as the target variable, describe how you would build a machine learning model to predict whether or not a student participates in the program.

Bonus: Implement the model you described in Python. Pseudocode is acceptable.

Table 4
Relationship between baseline characteristics and program takeup

	Aflatoun (1)	HMB (2)	Combined (3)
Student Has Money Saved	0.0115 (0.0464)	0.156** (0.0438)	0.0690* (0.0353)
Savings Attitude Index	0.0110 (0.0191)	0.00534 (0.0350)	0.00728 (0.0167)
Home Savings Support Index	-0.0272 (0.0271)	-0.0232 (0.0369)	-0.0226 (0.0214)
Work Index	-0.0104 (0.0166)	0.0219 (0.0392)	0.00827 (0.0226)
Risk Preference Index	0.0162 (0.0187)	-0.0106 (0.0227)	0.00584 (0.0168)
Time Preference Index	-0.00937 (0.0215)	0.000660 (0.0416)	-0.00522 (0.0220)
Financial Literacy Index	0.0166 (0.0130)	0.0909** (0.0346)	0.0492** (0.0206)
Expenditures on Temptation Goods Index	0.0255 (0.0300)	0.0146 (0.0369)	0.0334 (0.0219)
Expenditures on Self Index	0.0300 (0.0282)	0.0176*** (0.00373)	0.0174*** (0.00395)
Academic Performance Index	0.0123 (0.0106)	-0.00469 (0.0392)	0.00154 (0.0176)
Female	0.109** (0.0555)	0.110* (0.0525)	0.118** (0.0409)
Age	0.00464 (0.0163)	-0.0364 (0.0218)	-0.0224 (0.0161)
Student ever repeated grade	0.0590 (0.0343)	0.0276 (0.0520)	0.0635* (0.0311)
Index of durable goods ownership	0.0164** (0.00624)	-0.0358 (0.0345)	-0.00487 (0.0212)
Mean of dependent variable	0.162	0.297	0.217
R-squared	388	271	659
Number of observations	0.0550	0.114	0.0621
Number of Schools	10	7	17

Takeup is defined as attendance at one or more Aflatoun or HMB club meetings, as indicated by the club roster sheet or attendance logs. Row variables are measured at baseline. Each column presents the results of an OLS regression of takeup on the row variables in the Aflatoun and/or HMB schools for which club rosters or attendance logs were collected. The index of durable goods ownership is constructed as the first principal component of ownership of a mobile phone, television, radio, mattress, bicycle, car, motorcycle, stove, fridge, three types of livestock, school uniform, and shoes for the student. Standard errors clustered at the school level, in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5. In the paper, the authors state:

...our limited attendance data highlight the importance of collecting monitoring and management data as part of impact evaluations (see Gugerty, Karlan, & Welsh, 2016). In retrospect, additional monitoring data would have helped to provide a richer understanding of the programs' functioning and the mechanisms underlying the observed results, both positive and null. (pg. 72)

Assume that Aflatoun approaches you to ask advice about setting up the data engineering to monitor participation in savings clubs. How would you recommend setting up the data collection and data engineering to create an ongoing monitoring system for Aflatoun's savings clubs? Assume that Aflatoun wants weekly reporting on savings groups participation and savings amounts.